

Recognizing $\mathrm{SL}_2(7)$ by its order and conjugacy class sizes

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190
xiaojianzeng@foxmail.com

Abstract

Let $\mathrm{cs}(H)$ be the set of conjugacy class sizes of a finite group H , including 1 and discarding multiplicities. We prove that $|H| = 336$ and $\mathrm{cs}(H) = \{1, 24, 42, 56\}$ imply $H \cong \mathrm{SL}_2(7)$. An elementary class-equation argument forces $|Z(H)| = 2$ and shows that the only normal subgroups are 1, $Z(H)$, and H ; hence H is quasisimple. The final identification uses the complete list of 228 groups of order 336 in SmallGrp 1.5.4. Two exact computations retain only catalogue index 114, and a matrix certificate verifies the surviving isomorphism on all 112 896 ordered products. Catalogue completeness and GAP group arithmetic are explicit dependencies. The result settles this fixed-group recognition assertion; the broader existential question for quasisimple groups is unchanged.

1 The recognition statement and its context

For a finite group H , write $x^H = \{h^{-1}xh : h \in H\}$ and

$$\mathrm{cs}(H) = \{|x^H| : x \in H\}.$$

Thus equal sets need not have equal class multiplicities. Put $S = \mathrm{SL}_2(\mathbb{F}_7)$, also denoted $\mathrm{SL}_2(7)$.

Theorem 1. *For every finite group H ,*

$$|H| = 336, \quad \mathrm{cs}(H) = \mathrm{cs}(S) \implies H \cong S.$$

Equivalently, up to isomorphism S is the unique group of order 336 with class-size set $\{1, 24, 42, 56\}$.

Neither perfectness nor a condition on the center is assumed. These properties follow from the numerical hypotheses in Lemma 3. The finite identification in Section 3 then completes the proof using the published SmallGroups catalogue.

Vasil'ev [1, Theorem 2.3 and Problem 3.7] discusses recognition by group order together with the class-size set for simple and quasisimple groups. His Problem 3.7 asks whether some quasisimple group fails to be determined by these data. Theorem 1 excludes S as such a witness; it does not answer that existential question. Gorshkov [2, Theorem 1] proves recognition of $\mathrm{SL}_2(5)$ by its class-size set up to an abelian direct factor. Akhlaghi et al. [3] prove the corresponding statement for $\mathrm{PSL}_2(q)$, $q > 3$. The projective target at $q = 7$ has class-size set $\{1, 21, 24, 42, 56\}$, so its stated hypothesis differs from ours.

There is also a closer structural precedent in the F -group classification recorded in [3, Lemma 2.6]. An F -group is a group in which containment between centralizers of noncentral elements forces equality. Our noncentral centralizer orders 6, 8, 14 are pairwise incomparable

under divisibility, giving this property. Together with the perfectness proved below, that classification offers an alternative identification route. We retain the explicit catalogue argument and claim no advance in classification methods. The contribution is the fixed recognition instance, its elementary reduction to quasisimplicity, and the replayable finite identification. A bounded examination of these primary statements found no equivalent stated fixed-group theorem; this is not a claim of priority. As broad adjacent finite-group context, Zeng [6] studies generation of equal-sign fiber products of symmetric groups.

2 The class-size set and the structural reduction

Lemma 2. *The group S has order 336 and $\text{cs}(S) = \{1, 24, 42, 56\}$.*

Proof. There are $7^2 - 1 = 48$ possible nonzero first columns of a determinant-one matrix. For each, the determinant condition is a nonzero linear equation in the second column, with seven solutions. Hence $|S| = 336$.

A nonscalar matrix $A \in S$ admits a vector v for which v, Av form a basis. Indeed, otherwise every one-dimensional subspace would be invariant; applying this to two independent vectors and their sum would make A scalar. Any matrix commuting with A is determined by its value on v , and is therefore a linear combination of I and A . Thus the commuting algebra is $\mathbb{F}_7[A]$, and $C_S(A)$ consists of its determinant-one elements. There are three possibilities.

1. If the characteristic polynomial has a repeated root, that root is 1 or -1 . Write $A = \lambda I + N$ with $N^2 = 0 \neq N$. Its commuting algebra consists of the matrices $aI + bN$, of determinant a^2 . Exactly $2 \cdot 7 = 14$ have determinant one.
2. If it has two distinct roots in $\mathbb{F}_7$, the commuting matrices are diagonal in an eigenbasis. The determinant-one diagonal matrices number 6.
3. If it is irreducible, the commuting algebra is $\mathbb{F}_{49}$, and determinant is the norm to $\mathbb{F}_7$. Its kernel on the multiplicative group has order $(49 - 1)/(7 - 1) = 8$.

These cases occur respectively for

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}, \quad \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

For the last matrix, $X^2 + 1$ is irreducible over $\mathbb{F}_7$. The scalar matrices in S are I and $-I$. By orbit-stabilizer, the class sizes are exactly

$$\{1, 336/14, 336/6, 336/8\} = \{1, 24, 42, 56\}.$$

□

Lemma 3. *If $|H| = 336$ and $\text{cs}(H) = \{1, 24, 42, 56\}$, then $|Z(H)| = 2$ and the only normal subgroups of H are 1, $Z(H)$, and H . Consequently H is perfect and $H/Z(H)$ is nonabelian simple of order 168.*

Proof. Every centralizer contains $Z(H)$, so

$$|Z(H)| \mid \gcd(336/24, 336/42, 336/56) = 2.$$

All noncentral classes have even size. The class equation and the even order of H therefore force $|Z(H)|$ to be even, whence $|Z(H)| = 2$.

Suppose N is a proper normal subgroup not contained in $Z(H)$. As a union of H -conjugacy classes, its order d satisfies

$$d = z + 24a + 42b + 56c, \quad z = |N \cap Z(H)| \in \{1, 2\}, \quad (1)$$

where a, b, c are nonnegative integers and $a + b + c > 0$. In particular $d \geq 25$. Lagrange's theorem gives the complete list

$$d \in \{28, 42, 48, 56, 84, 112, 168\}.$$

All these orders are even, so (1) forces $z = 2$.

For $d = 48$, necessarily $c = 0$, and $46 = 24a + 42b$ is impossible modulo 6. Every other listed d is divisible by 7. Reduction of (1) modulo 7 yields $3a \equiv -2 \pmod{7}$, or $a \equiv 4 \pmod{7}$. Hence $a \geq 4$ and $d \geq 98$, leaving $d = 112$ or 168 . Since $a \leq \lfloor (d-2)/24 \rfloor \leq 6$, in both cases $a = 4$. At $d = 112$ the remaining equation is $42b + 56c = 14$, which is impossible. At $d = 168$ it is $3b + 4c = 5$: the only possibilities $c = 0, 1$ require $3b = 5, 1$, respectively. These are also impossible.

Thus every proper normal subgroup lies in the center of order 2. Normal-subgroup correspondence makes $H/Z(H)$ simple of order 168. A finite abelian simple group has prime order, so this quotient is nonabelian. The derived subgroup H' cannot lie in $Z(H)$, since that would make the quotient abelian. The normal-subgroup list now forces $H' = H$. $\square$

A perfect group with nonabelian simple central quotient is called quasisimple. Lemma 3 establishes this conclusion without using the group catalogue or the class multiplicities of S .

3 Complete finite identification

The classification input is the complete isomorphism-type coverage at order 336 in the SmallGroups library [4, Sections 1.1, 1.3 and Chapter 2]. The installed versions used for the computations were GAP 4.15.1 [5] and SmallGrp 1.5.4. There are exactly 228 catalogue entries of this order. Thus every finite group of order 336 is isomorphic to

$$G_i = \text{SmallGroup}(336, i), \quad 1 \leq i \leq 228.$$

This uses the catalogue's completeness, including its nonsolvable entries.

The first sweep computes $\text{cs}(G_i)$ by GAP's conjugacy-class routine. The second enumerates all 336 distinct elements of each G_i and, for each element x , computes

$$c_i(x) = \sum_{y \in G_i} \mathbf{1}_{xy=yx}, \quad D_i = \{336/c_i(x) : x \in G_i\}. \quad (2)$$

Every count is positive and divides 336. Since $c_i(x) = |C_{G_i}(x)|$, orbit-stabilizer gives $D_i = \text{cs}(G_i)$. The second calculation uses only element enumeration, multiplication, and equality for this invariant. Its core is the following GAP code; the accompanying script also checks coverage, records all counts, and exports the survivor certificate.

```
target := [1,24,42,56];; hits := [];;
for i in [1..228] do
  g := SmallGroup(336,i);
  el := AsList(g);
  counts := List(el, x -> Number(el, y -> x*y = y*x));
  if Set(counts, c -> 336/c) = target then
    Add(hits,i);
  fi;
od;
# hits = [114]
```

The two sweeps agree at every index. The second covers $228 \cdot 336 = 76\,608$ elements and $228 \cdot 336^2 = 25\,740\,288$ ordered pairs. No solvability filter or sampling is used. Both yield exactly

$$\{i : D_i = \{1, 24, 42, 56\}\} = \{114\}. \quad (3)$$

To identify G_{114} , the exporter lists its elements $g_0, \dots, g_{335}$, its full multiplication table T , and the images $M_0, \dots, M_{335}$ of a GAP-constructed isomorphism into S . An integer Python checker enumerates all $7^4 = 2401$ tuples (a, b, c, d) modulo 7 and retains exactly the 336 with $ad - bc = 1$. It verifies that the image list is a bijection onto this set, that every table index is in $\{0, \dots, 335\}$, and that

$$M_{T(i,j)} = M_i M_j \pmod{7} \quad (0 \leq i, j < 336). \quad (4)$$

All $336^2 = 112\,896$ products satisfy (4). Thus the supplied bijection is an isomorphism. The connection between the table and catalogue index 114 is supplied by the inspected GAP exporter.

Completion of the proof of Theorem 1. By Lemma 2, the hypotheses specify the class-size set $\{1, 24, 42, 56\}$. Catalogue completeness gives $H \cong G_i$ for some $1 \leq i \leq 228$. Equation (3) forces $i = 114$, and the bijective multiplicative certificate (4) gives $G_{114} \cong S$. $\square$

As a consistency check, direct conjugation of the integer matrices gives the following distribution. These multiplicities are computed conclusions and are never imposed on the catalogue search.

Class size	1	24	42	56
Number of classes	2	4	3	2
Centralizer order	336	14	8	6

The weighted sum is $2 + 4 \cdot 24 + 3 \cdot 42 + 2 \cdot 56 = 336$.

4 Reproducibility and limitations

The accompanying supplement contains both GAP scripts and complete outputs, the 336-element table and matrix certificate, and a Python verifier using only standard-library integer arithmetic. The verifier requires each index $1, \dots, 228$ exactly once, reconstructs every class-size set from the saved commuting counts, compares both sweeps, checks the matrix target, and tests every certificate product. It also exhausts (1) over all divisors of 336 as an additional check on Lemma 3. The supplement's README gives commands for verification and for regeneration in a separate working copy.

A separate computational review replayed the full commuting sweep and regenerated the table. A separately written Python checker verified the frozen and regenerated data, reconstructed matrix conjugation orbits, and checked all products in both certificates. Both GAP sweeps and the replay share the catalogue and GAP group arithmetic. The integer certificate check independently tests the serialized map and matrix operations; it does not establish catalogue completeness or independently classify groups of order 336. No proof-assistant formalization is claimed.

The result retains the order hypothesis and concerns only $\mathrm{SL}_2(7)$. The broader quasisimple recognition problem remains unresolved by this work. The literature comparison is bounded, and the F -group classification provides relevant prior structural machinery.

Computational and AI assistance. OpenAI Codex assisted with the mathematical argument, literature screening, GAP and Python code, manuscript preparation, and a separate mathematical review pass. Exact outputs and product certificates, together with the stated classification dependency, support the computational steps. The AI review is not external peer review or proof-assistant verification.

References

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